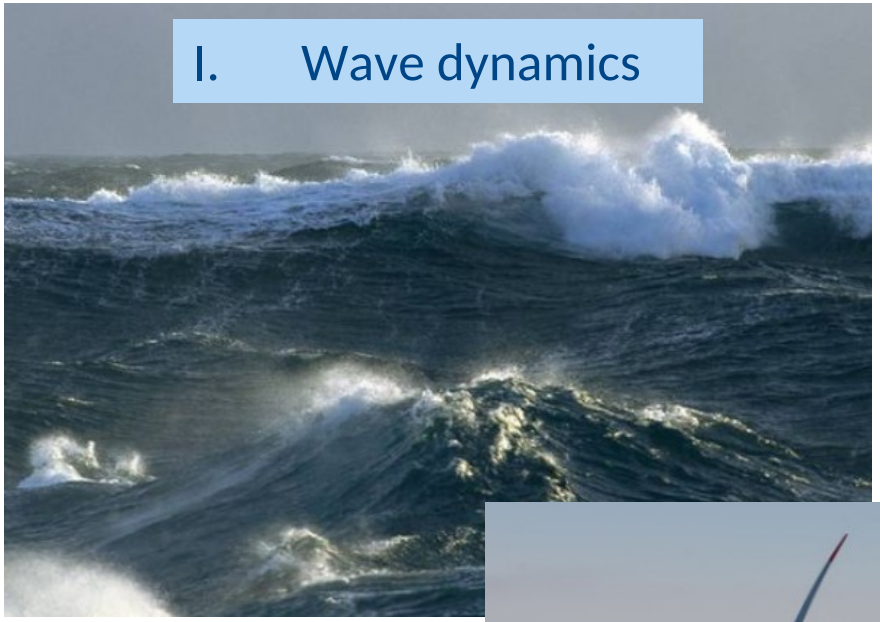
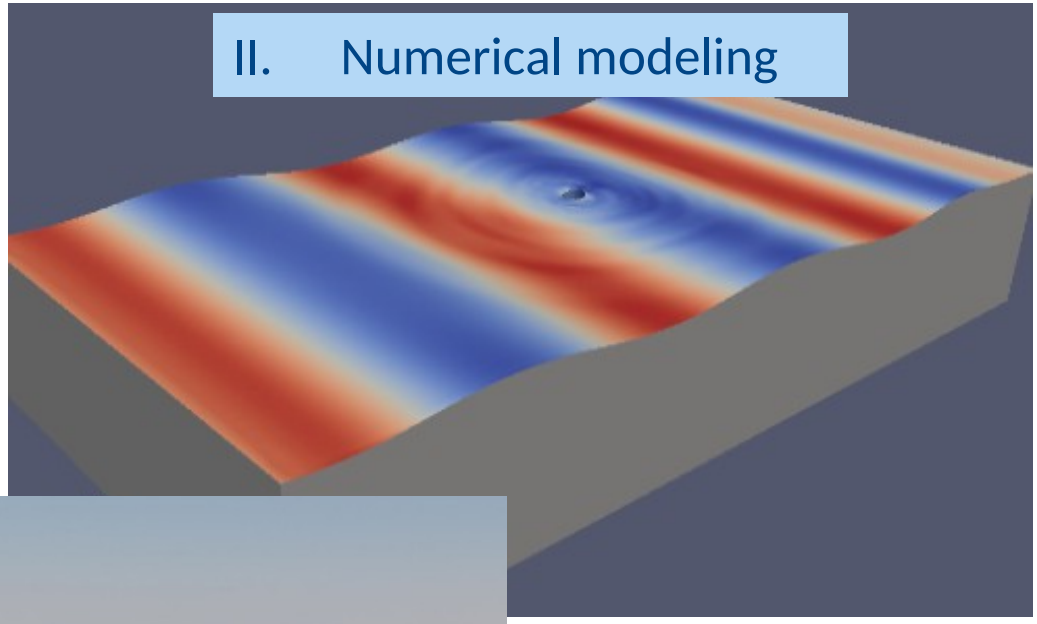
SSWPOWE Class project





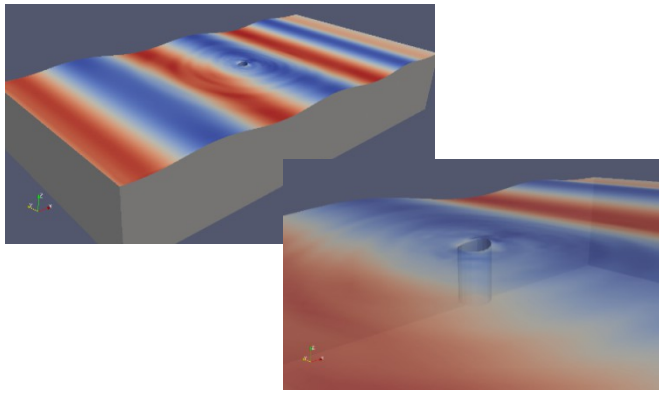


Class project



I. Wave dynamics

- Characterizing the wave characteristics at the selected study site and estimating the wave conditions to simulate;



II. Numerical modeling

- Applying a numerical model to propagate waves from deep to shallow water conditions to estimate the wave forces on a structure;



III. Wave-structure interactions

- Estimating the movement of the wave turbine as well as the design criteria for the anchoring system.

Class project

Introduction and site selection (1 class)

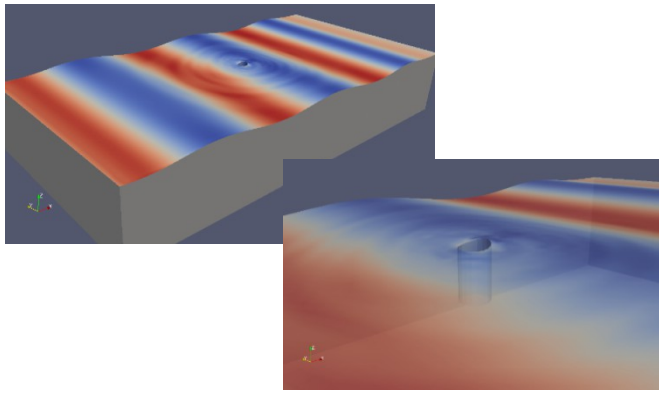
I. Wave dynamics (2 classes)

- Characterizing the wave characteristics at the selected study site and estimating theextreme wave conditions;



II. Numerical modeling (2 classes)

- Applying a numerical model to propagate waves from deep to shallow water conditions to estimate the wave forces on a structure;



III. Wave-structure interactions (2 classes)

- Estimating the movement of the wave turbine as well as the design criteria for the anchoring system.



Finish analysis and prepare presentation (1 class)

Course organization

I. Wave dynamics
II. Numerical modeling
III. Wave-structure interactions

Structure:

Lectures
In-class exercises
Project (via TD's)
Mini presentations

Grading:

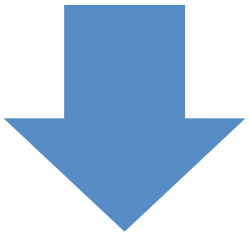
Class participation – 20%
Mini presentations – 20%
Project reports – 30%
Final project presentation – 30%

Reports = individual
Presentation = group evaluation

Offshore wind farm project

Characterization, modeling and design of a wind farm

- **Applying theories presented in class**
- **Wave data analysis**
- **Running simple models and comparing results**
- **Modeling wind turbine stability**



Comprehension of course material *(group analyses)*

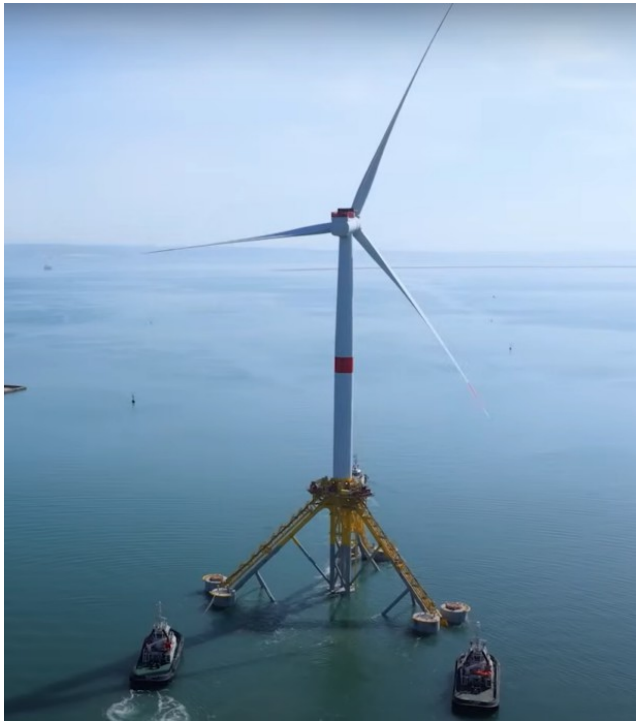
Synthesizing results *(individual reports)*

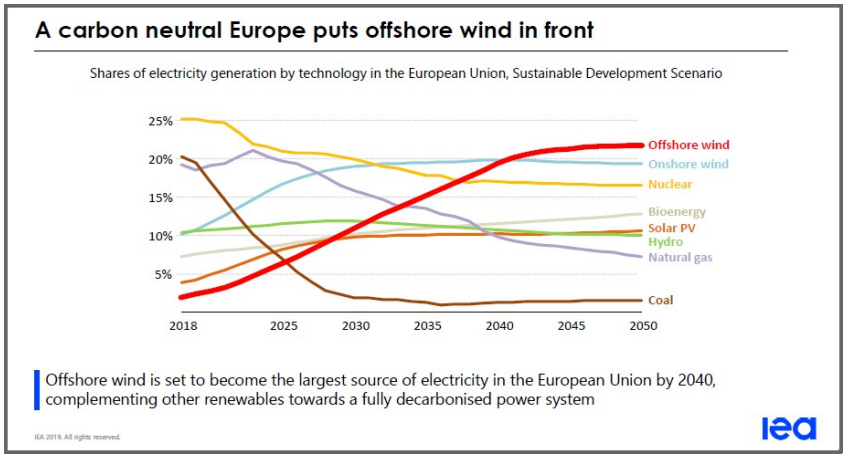
Clear presentation of your work *(final oral presentation)*

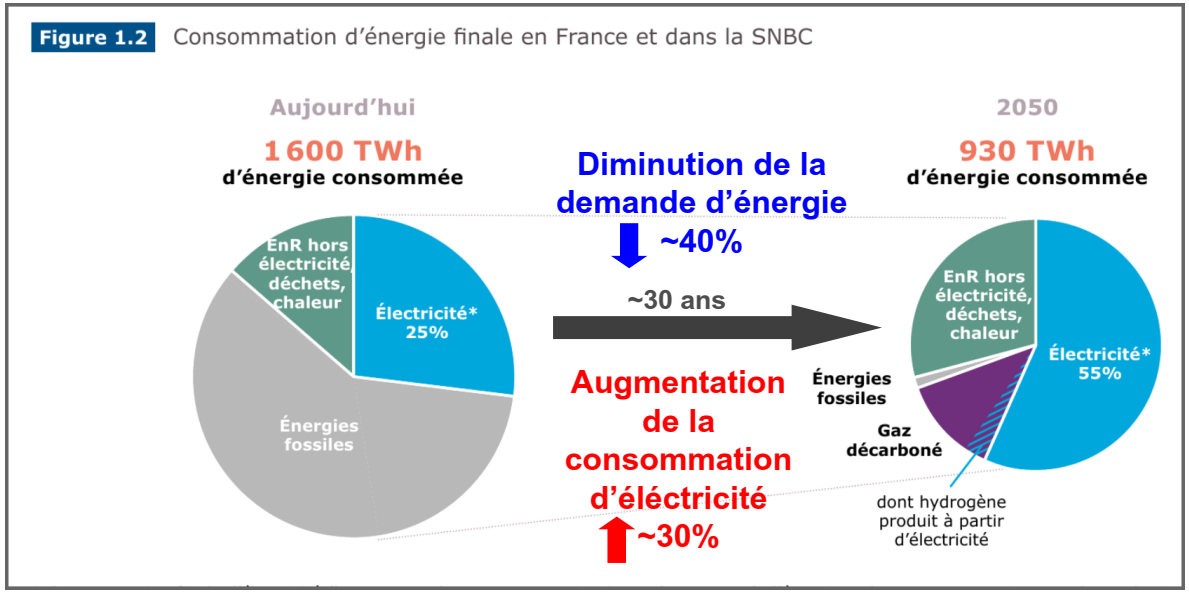
Wind turbine development

France aims to be carbon neutral by 2050

- offshore wind is the 2nd most important contribution
- goal : 50 GW installed in 2050

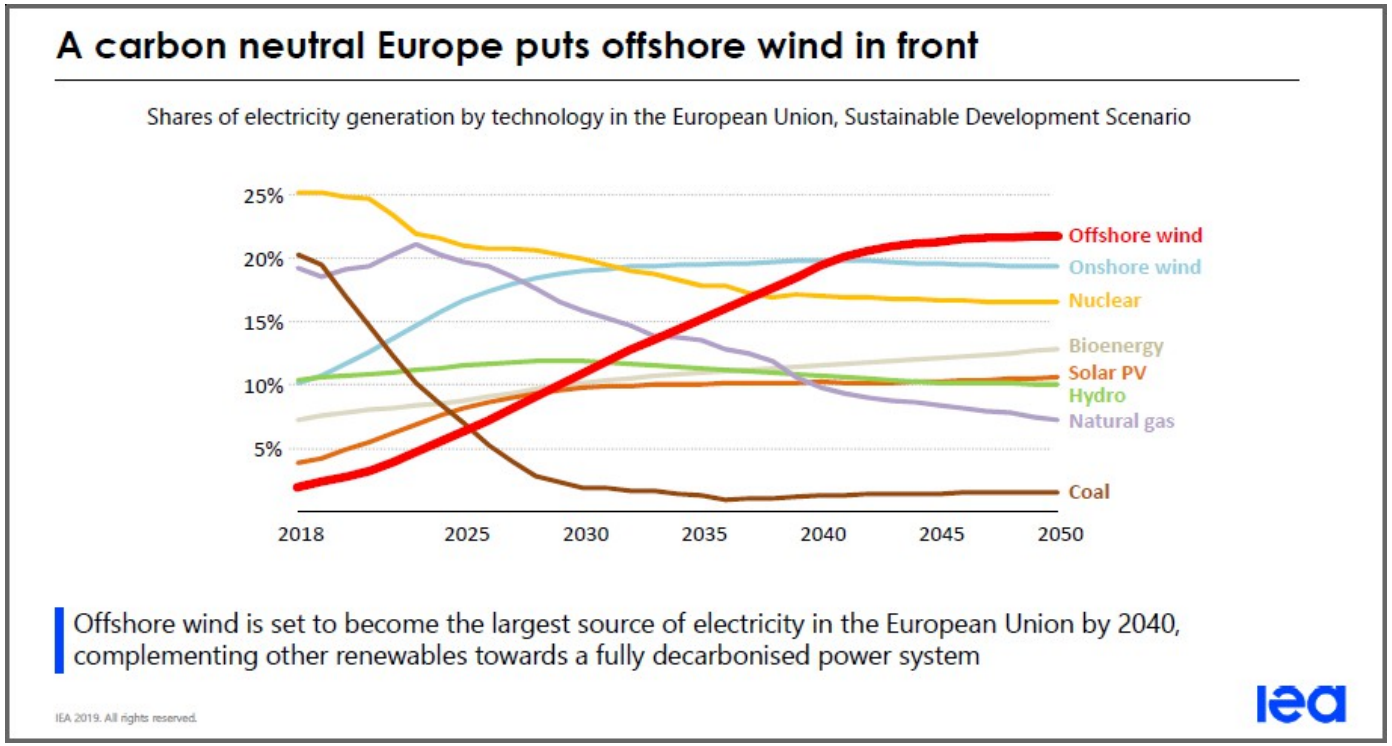






(Futurs énergétiques 2050, Rapport complet, Février 2022, RTE)

Wind turbine development



Offshore wind

Global annual offshore wind potential: 420 000 TWh

= 18 times the global electricity demand (11 times what is predicted in 2040)

Europe annual offshore wind potentia : 34 000 TWh

= with a demand of <3000 TWh

(IIFP Energies Nouvelles, d'après l'IEA)

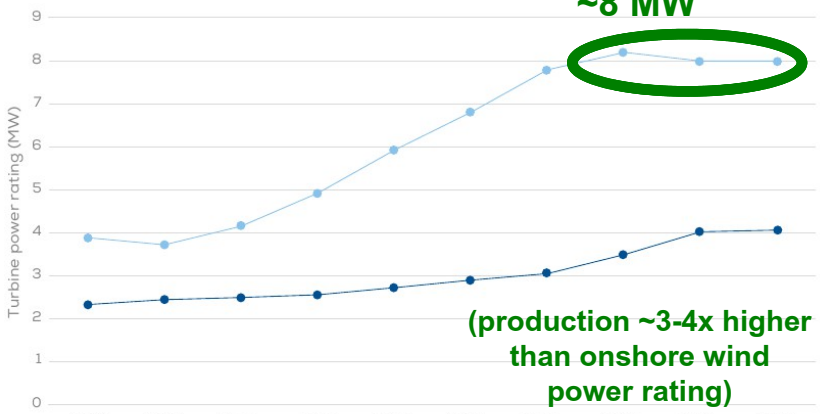
Wind turbine development

In Europe...

- 30 GW installed offshore (end of 2022)
- with ~2-3 GW/year
- primaily (~100%) fixed wind turbines

FIGURE 15. Average power rating of installed turbines in Europe, 2013-22

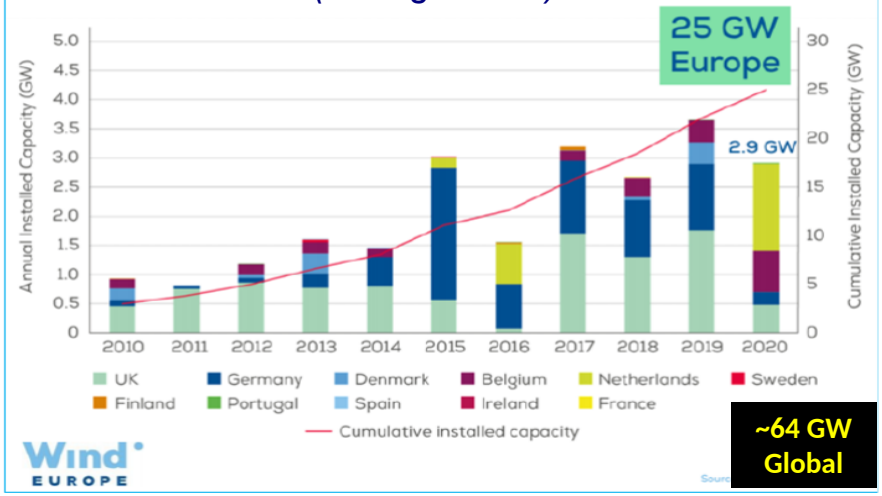
offshore wind turbines ~8 MW



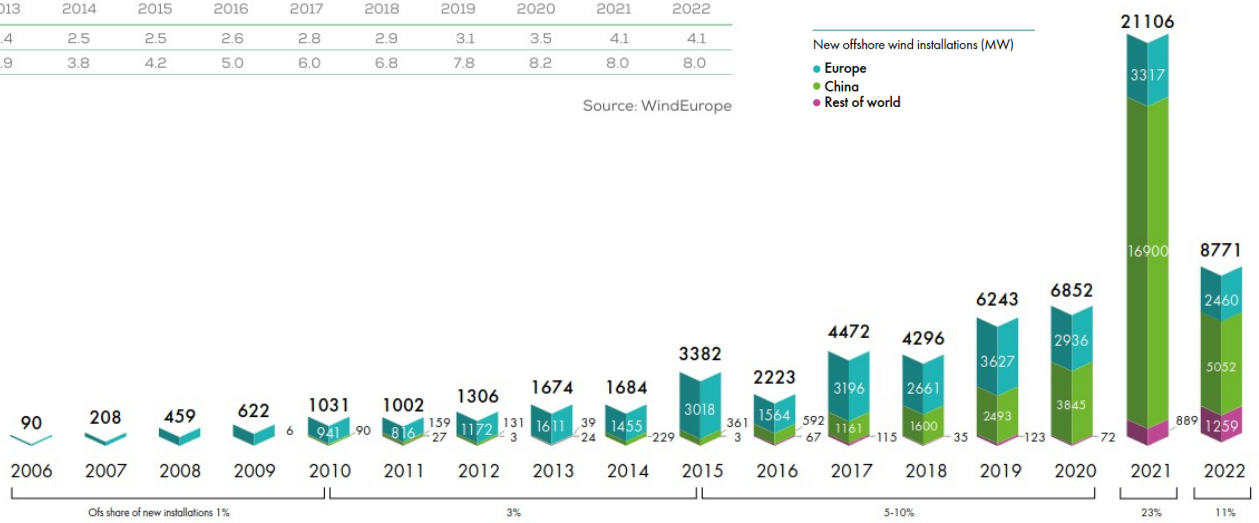
(production ~3-4x higher than onshore wind power rating)

Source: WindEurope

Global annual offshore installations *(through 2020)*



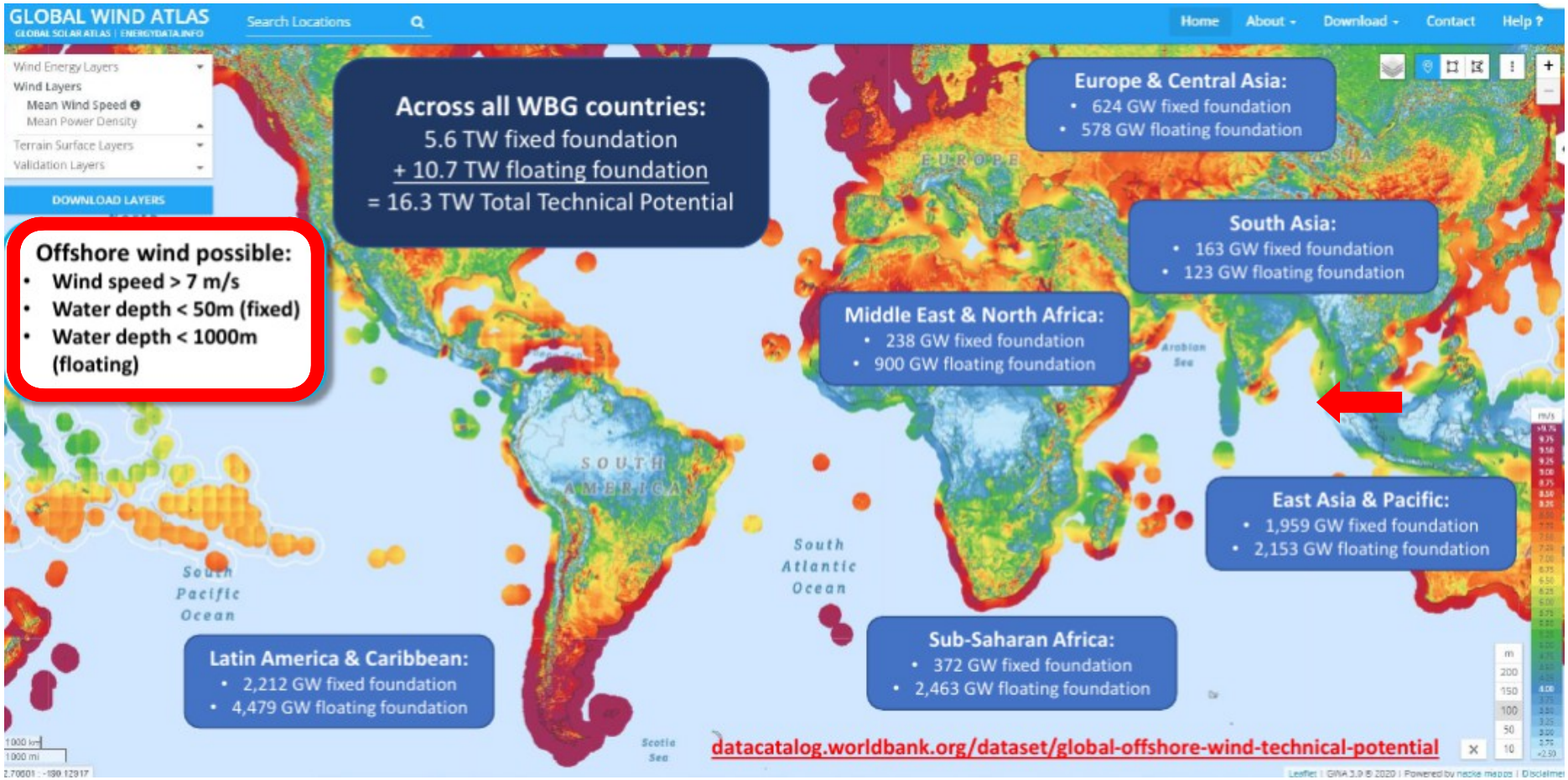
~64 GW Global



Offshore wind is the fastest growing renewable energy resource (+19% between 2010 et 2022)

(Jensen, AEGIR, 2022)

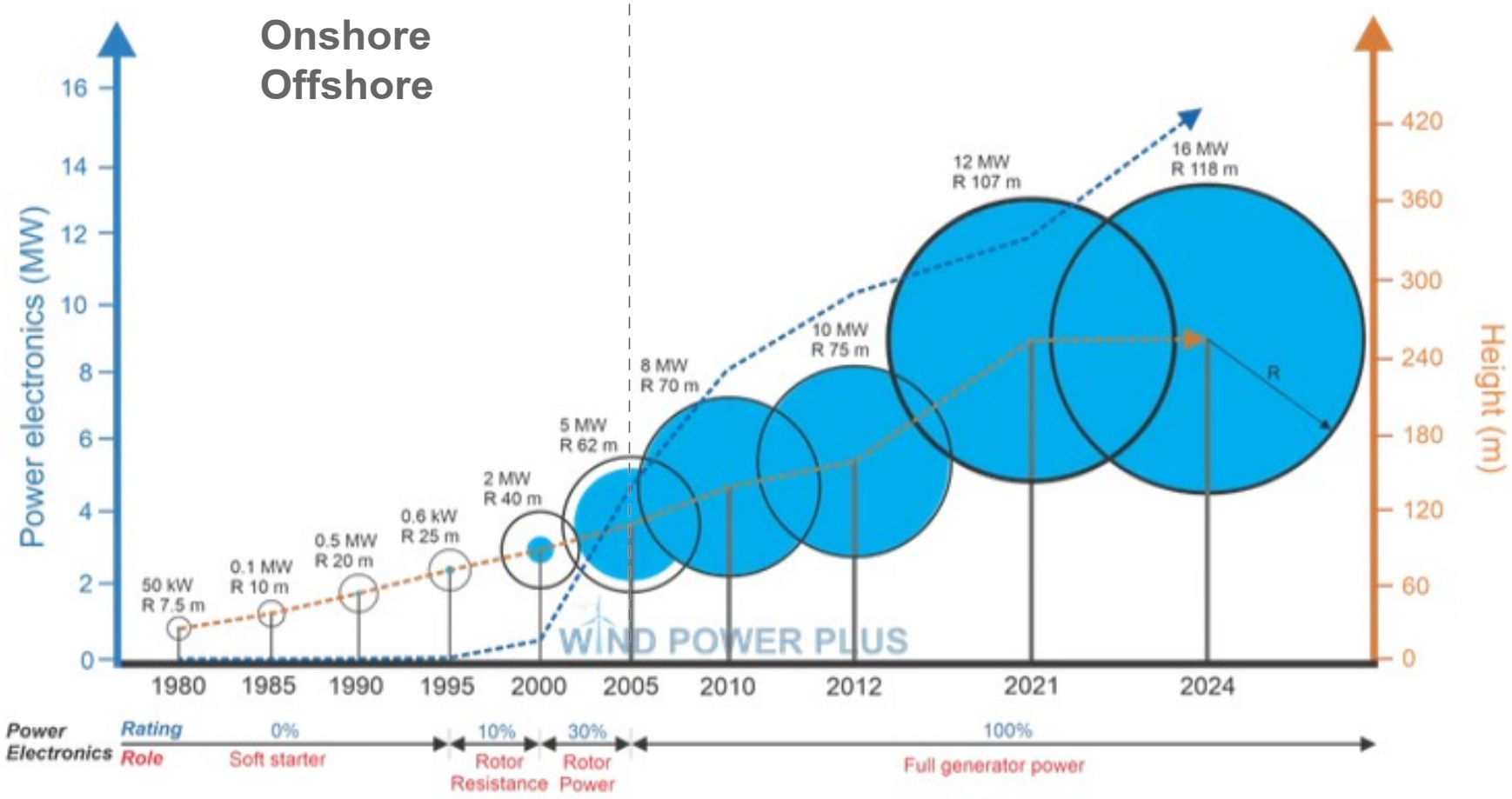
Wind turbine development



(ESMAP, World Bank Group Offshore Wind Development Program, Update)

- <200 km coast
- isolated areas excluded (<10 km^2)

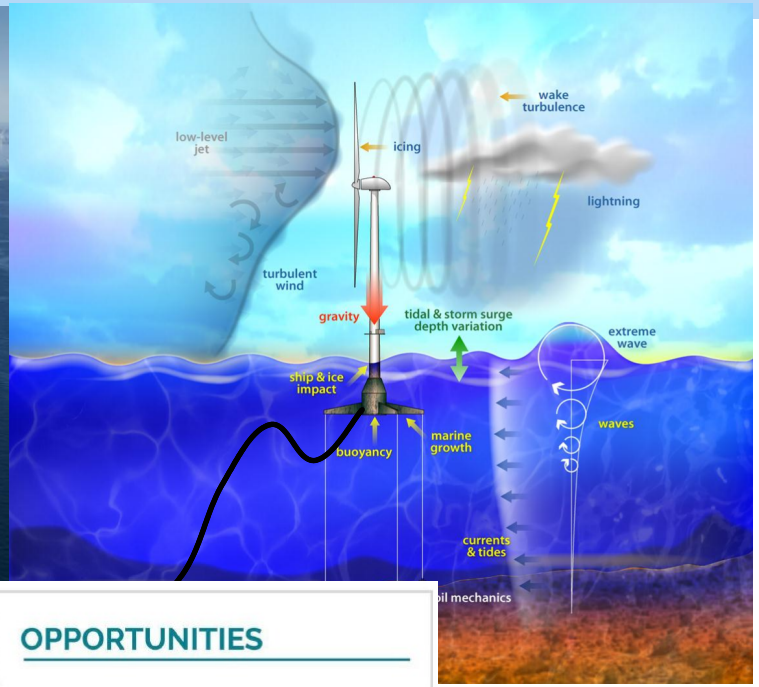
Wind turbine development

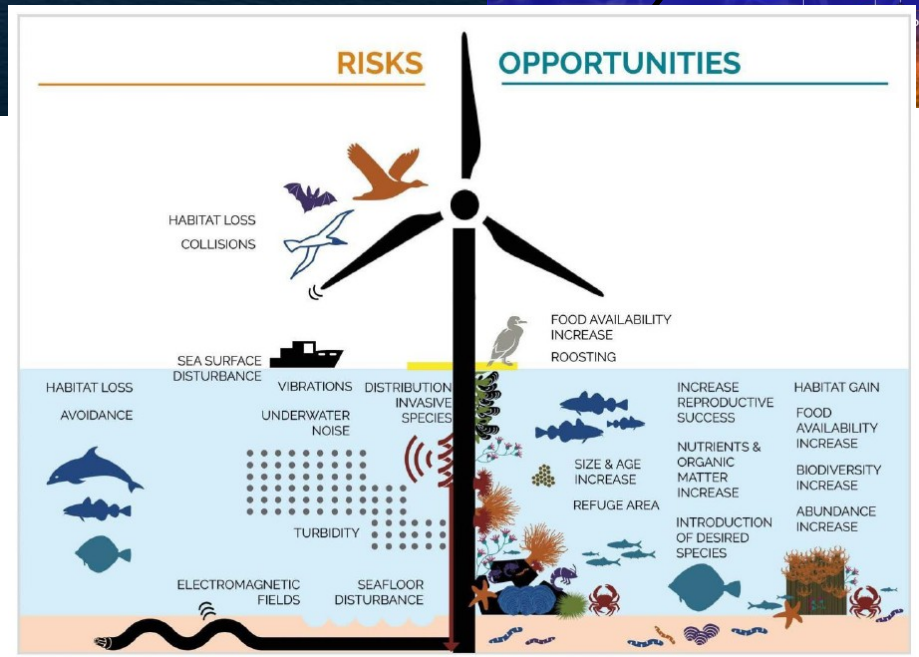


(https://windpowerplus.com/history-of-wind-energy/)

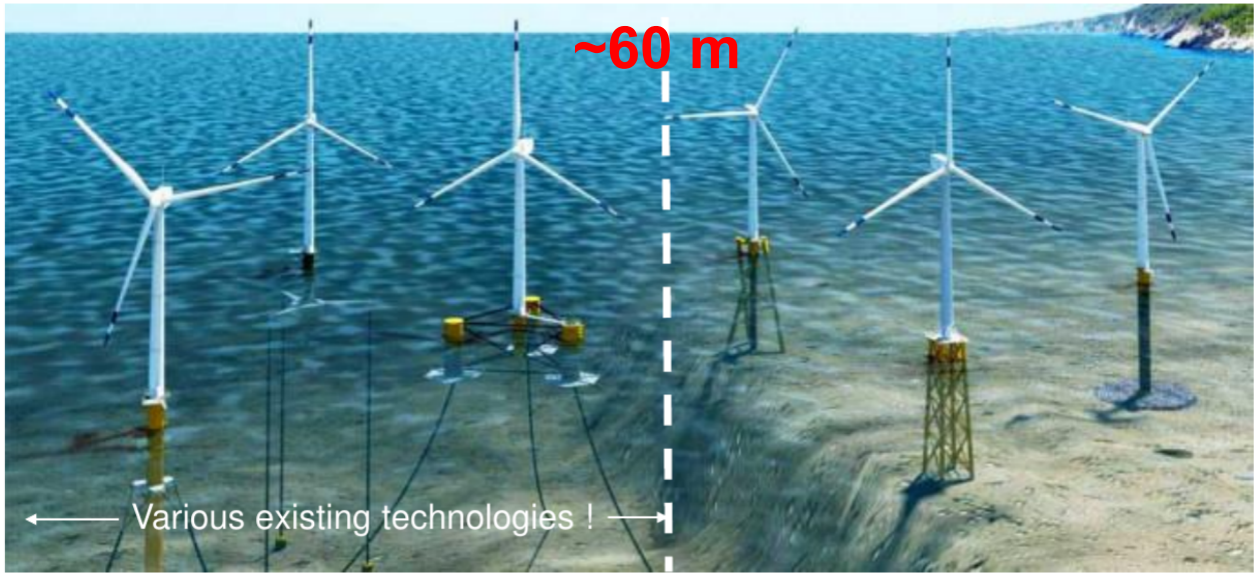
Wind turbine development







Wind turbine development



COUNTRY / REGION	SHARE OF OFFSHORE WIND RESOURCE IN +60m DEPTH	POTENTIAL FOR FLOATING WIND CAPACITY
Europe	80%	4,000 GW
USA	60%	2,450 GW
Japan	80%	500 GW
Taiwan	-	90 GW

Sources: MOFA and Carbon Trust[2]

(WindEurope, 2017)

Available resource

- 80% of the offshore wind resource in Europe is in zones >60 m water depth

- Floating Offshore Wind (FOW) turbines are less expensive than Bottom Fixed Offshore Wind (BFOW) turbines for water depths >60 m

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Liberté
Égalité
Fraternité

Eoliennes en mer en France



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Liberté
Égalité
Fraternité

Eoliennes en mer en France



https://www.eoliennesenmer.fr/

Site selection

1. **Wind resource**: financially viable with average wind speed at the hub height >6 m/s, and safe investment if >8.5 m s−1
2. **Marine aspects**: water depth, wave conditions, current and tide data, sediment transport, identification of scour-related issues (protection needed ?)
3. **Environmental impact**: environmental data and potential impacts, including biological (marine mammals, sea birds, effects on flora and fauna) noise during construction and operation, and electro-magnetic field generated by subsea cable, marine archaeology (shipwrecks)
4. **Power export/grid connection**: length of submarine cable required, which is dependent on the turbine layout, substation location, export cable routing (landfall), risk assessment of buried cables, and the transformer options
5. **Economics**
6. **Navigation**: investigate whether there is a need for exclusion zone due to fishing or navigation or military operations. Cables connecting the wind farms and the export cables are buried to depths of 2–3 m to avoid risk of entanglement with nets. Navigation risk must be assessed.
7. **Consents and legislations**

Class project: 2 individual reports

Report I: wave analysis
Report II: wave modeling

Recommended structure:

- **Introduction and context**
- **Description of methods and model**
- **Results**
- **Discussion: preliminary study, what needs to be done next to go further?**
- **Conclusion**
- **References**

Class project: final group presentation

Final presentation: wind turbine design (including elements from Parts I and II as necessary)

Recommended structure:

- **Introduction and context**
- **Description of methods and model**
- **Results**
- **Discussion: preliminary study, what needs to be done next to go further?**
- **Conclusion**
- **References**

Project schedule

Due dates in red: reports (R) and final presentation (FP)

I. Wave dynamics	Project	Lecturer	Date	
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	25/09	
2. Linear wave theory	Characterizing offshore waves	M. Yates	09/10	
3. Nearshore wave propagation	Calculating extreme wave conditions	M. Yates	16/10	
4. Coastal hydrodynamics	**Student mini-presentations**	M. Yates	**06/11****	
II. Numerical wave modeling				
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	**23/10****	**R1**
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	13/11	
III. Wave-structure interactions				
7. Dynamics of a body in waves	Wave load estimation	J. Harris	20/11	
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	**04/12**	**R2**
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	11/12	
10. Final project presentations 14h - 18h15			***18/12***	**FP**

Getting started….

Backgroud research and site selection:

(in French) Eoliennes en mer en France:

https://www.eoliennesenmer.fr/

https://globalwindatlas.info/en/area/France



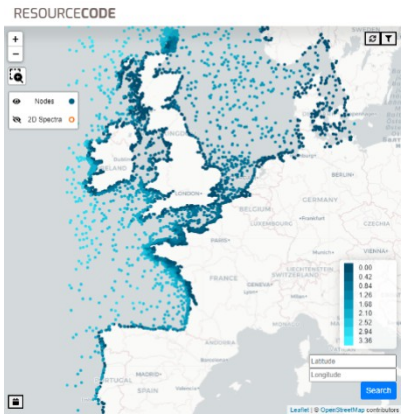
Availability of wave buoy observations via the Candhis network:

https://candhis.cerema.fr/index.php



Availability of long-term wave simulations via Copernicus:

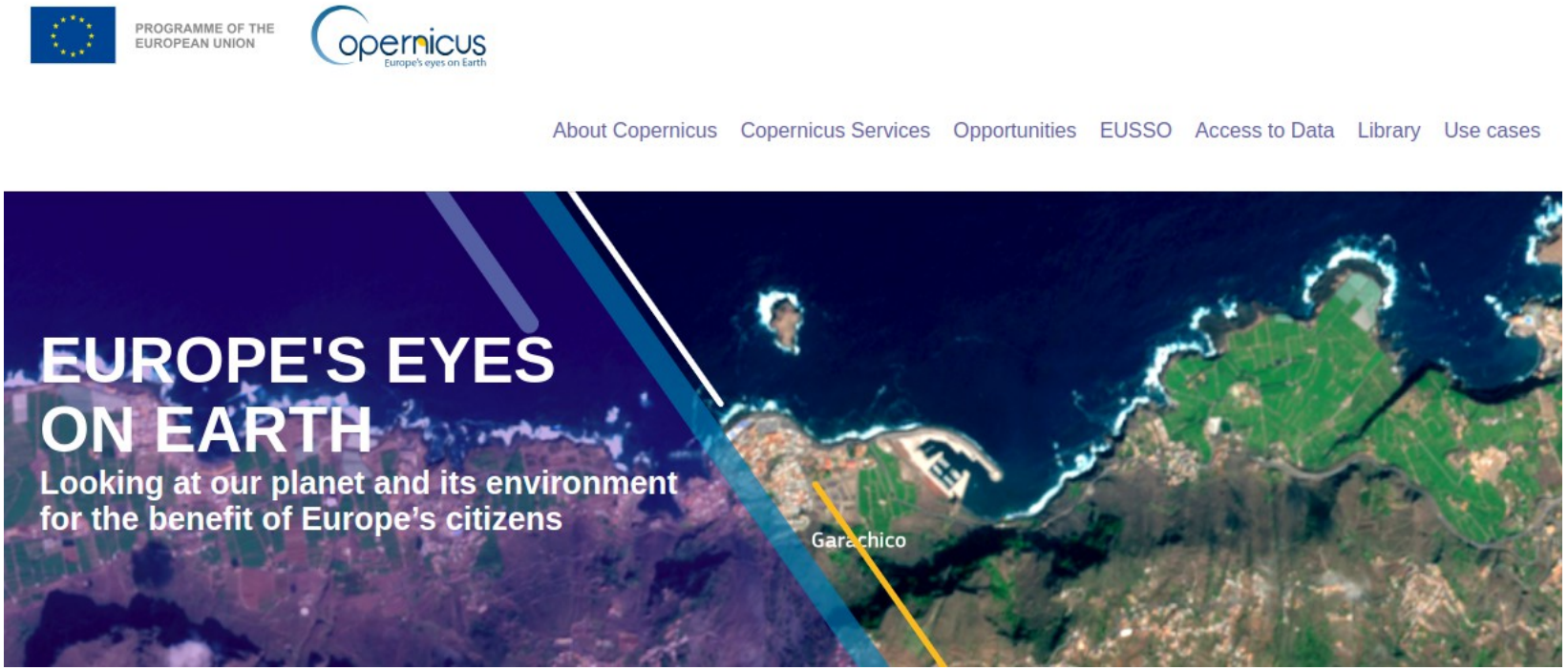
https://data.marine.copernicus.eu



Download wave observations

Many different wave data sources worldwide :

- Copernicus (European action with global data coverage)

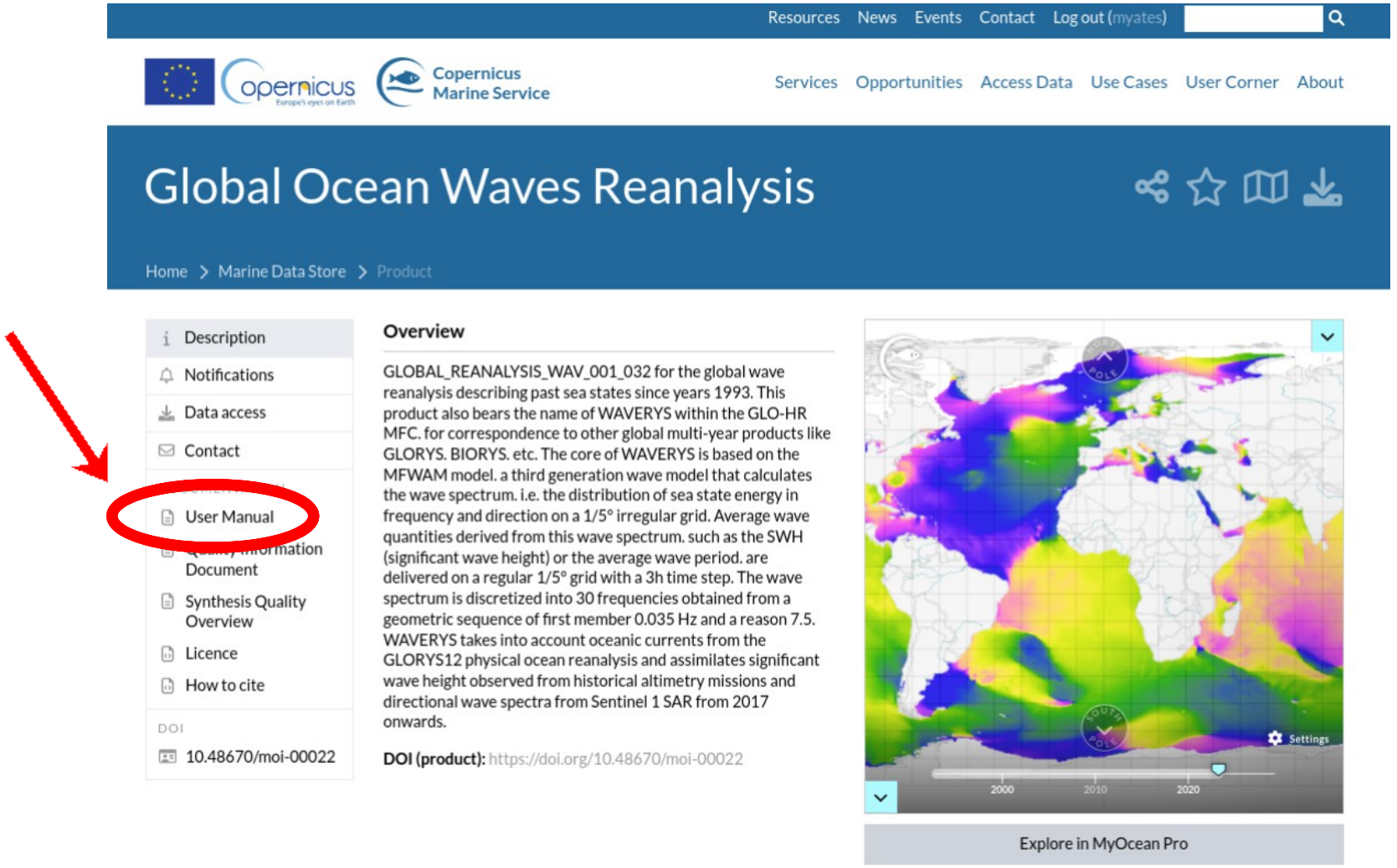


- among others...

Downloading wave data from Copernicus

Global Ocean Waves Reanalysis

https://data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_WAV_001_032/description



Downloading wave data from Copernicus

User Manual



The product is organized in 2 datasets:

- **cmems_mod_glo_wav_my_0.2deg_PT3H-i** containing 3-hourly instantaneous values for all the variables on multiyear period
- **cmems_mod_glo_wav_my_0.2deg-climatology_ P1M-m** containing monthly mean climatology values for sea surface wave significant height and sea surface wave mean period from variance spectral density second frequency moment on 1993-30/04/2019 period.
- **cmems_mod_glo_wav_myint_0.2deg_PT3H-i** containing 3-hourly instantaneous values for all the variables on interim period
- **cmems_mod_glo_wav_my_0.2deg_static** containing the bathymetry

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User Manual



The product is organized in 2 datasets:
- **cmems_mod_glo_wav_my_0.2deg_PT3H-i** containing 3-hourly instantaneous values

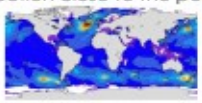
Product name	**GLOBAL_MULTIYEAR_WAV_001_032**
Geographical coverage	Global
Variables	**For cmems_mod_glo_wav_my_0.2deg_PT3H-i and cmems_mod_glo_wav_myint_0.2deg_PT3H-i:** Spectral significant wave height (Hm0) Spectral moments (-1,0) wave period (Tm-10) Spectral moments (0,2) wave period (Tm02) Wave period at spectral peak / peak period (Tp) Mean wave direction from (Mdir) Wave principal direction at spectral peak Stokes drift U Stokes drift V Spectral significant wind wave height Spectral moments (0,1) wind wave period Mean wind wave direction from Spectral significant primary swell wave height Spectral moments (0,1) primary swell wave period Mean primary swell wave direction from Spectral significant secondary swell wave height Spectral moments (0,1) secondary swell wave period Mean secondary swell wave direction from **For cmems_mod_glo_wav_my_0.2deg-climatology_ P1M-m:** Spectral significant wave height (Hm0) Spectral moments (0,2) wave period (Tm02)
	Analysis
Available time series	15th January 1993 to 30/04/2023 for multiyear period 01/05/2023 to M-4 for interim period
Temporal resolution	3-hourly instantaneous
Delivery mechanisms	Subsetter and FTP
Horizontal resolution	1/5 ° (equirectangular grid)
Number of vertical levels	surface only
Format	NetCDF CF1.6

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User Manual



The product is organized in 2 datasets:
- **cmems_mod_glo_wav_my_0.2deg_PT3H-i** containing 3-hourly instantaneous values

Product name	**GLOBAL_MULTIYEAR_WAV_001_032**
Geographical coverage	Global
Domain **Resolution and grid**	**GLOBAL (180°W-179.8°E ; 89.8°S – 89.8°N)** **1/5° ; regular grid ; 1800 x 899**
Geographic coverage	This product is global with dedicated projection and spatial resolution. It is defined on a standard collocated grid at 1/5 degree (approx. 20 km). The wave parameters are interpolated from the native grid model, which is irregular with increasing distance in the latitudinal direction close to the poles. 
Algorithm	MFWAM (ECWAM-IFS-38R2 source code)
Atmospheric forcings	ERA5 climatological winds and ice fraction: 3-hourly analysis
Current forcing	GLORYS12V1 current (GLOBAL_MULTIYEAR_PHY_001_030 Copernicus Marine product): 3-hourly analysis
Assimilation scheme	Optimal interpolation
Assimilated observations	Significant wave height (SWH) from altimeters ERS1, TOPEX/POSEIDON, ERS2, GFO, Jason 2 & 3, Envisat, Saral, Cryosat-2 and Sentinel-3A; Sentinel-3 and Sentinel-1-SAR spectrum
Initial conditions	No
Bathymetry	ETOPO2
Format	NetCDF CF1.6

Downloading wave data from Copernicus

User Manual



The product is organized in 2 datasets:
- **cmems_mod_glo_wav_my_0.2deg_PT3H-i** containing 3-hourly instantaneous values

Product name	**GLOBAL_MULTIYEAR_WAV_001_032**
Geographical coverage	Global
Domain **Resolution and grid**	**GLOBAL (180°W-179.8°E ; 89.8°S – 89.8°N)** **1/5° ; regular grid ; 1800 x 899**
Initial conditions	No
Bathymetry	ETOPO2
Format	NetCDF CF1.6

GLOBAL_REANALYSIS_WAV_001_032

cmems_mod_glo_wav_my_0.2deg_PT3H-i and **cmems_mod_glo_wav_myint_0.2deg_PT3H-i** contains instantaneous data

VHM0 [m]
Spectral significant wave height (Hm0)
sea_surface_wave_significant_height

VTM10 [s]
Spectral moments (-1,0) wave period (Tm-10)
sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_moment

VTM02 [s]
Spectral moments (0,2) wave period (Tm02)
sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment

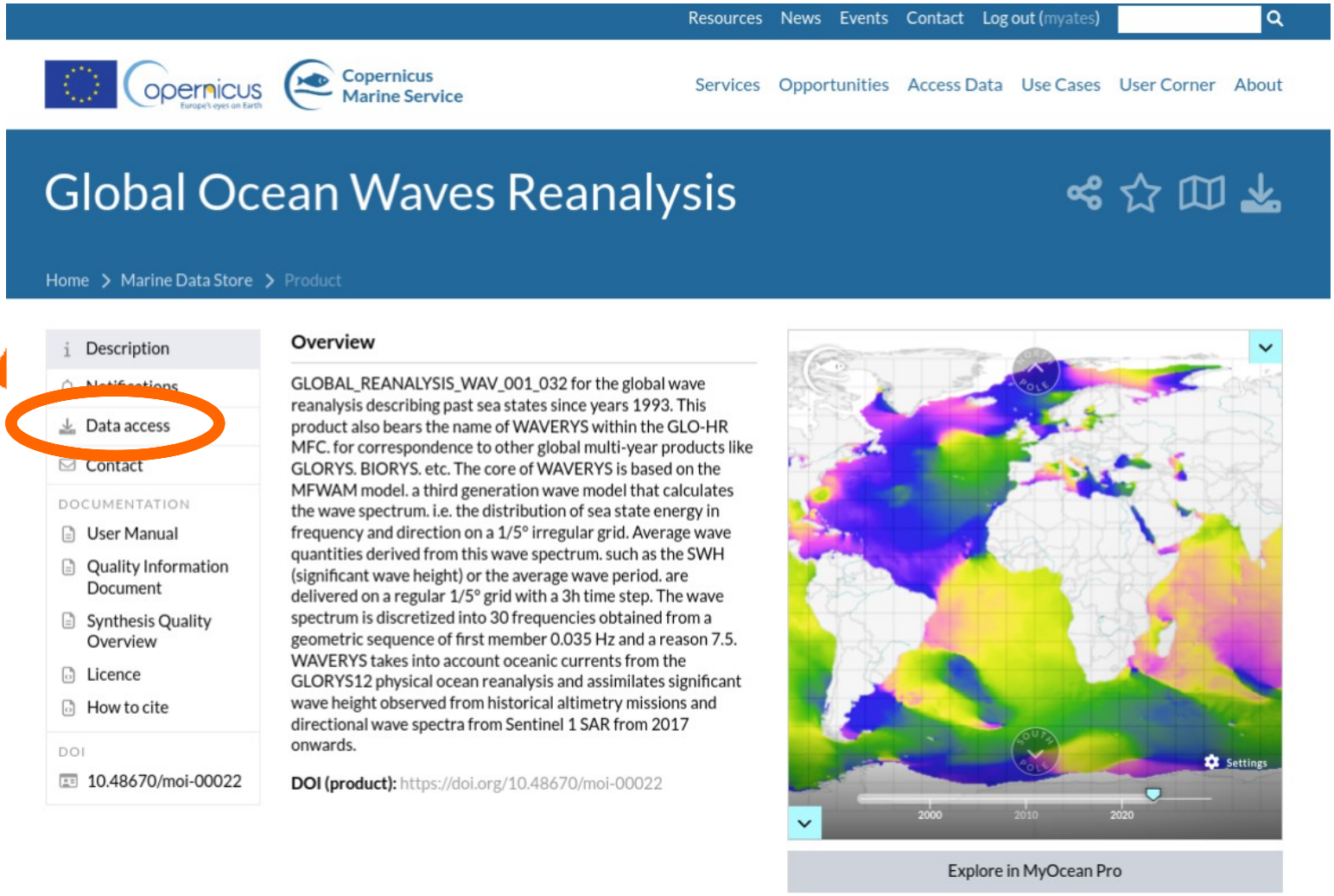
VTPK [s]
Wave period at spectral peak / peak period (Tp)
sea_surface_wave_period_at_variance_spectral_density_maximum

VMDR [degree]
Mean wave direction from (Mdir)
sea_surface_wave_from_direction

Downloading wave data from Copernicus

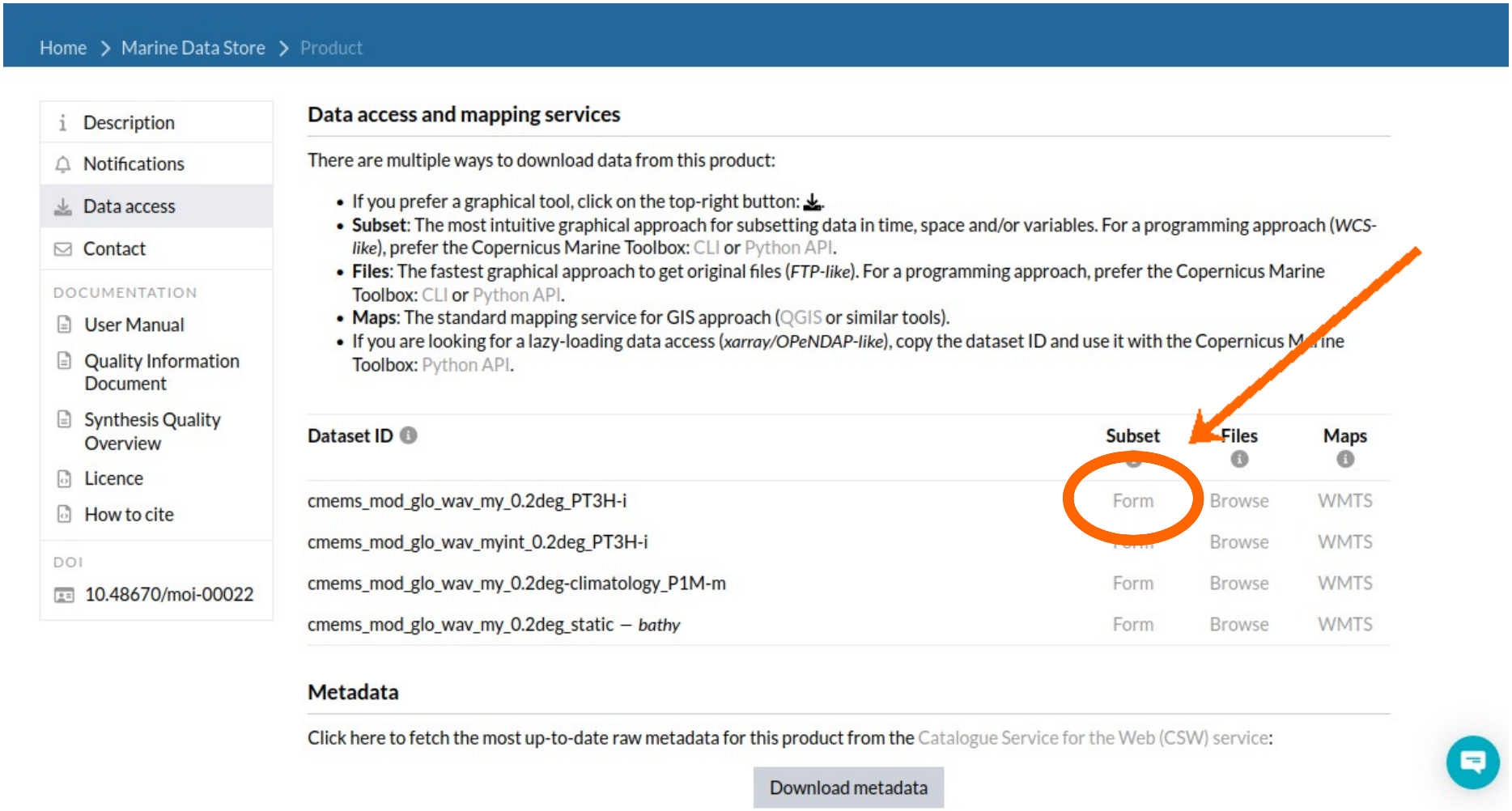
Global Ocean Waves Reanalysis

https://data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_WAV_001_032/description



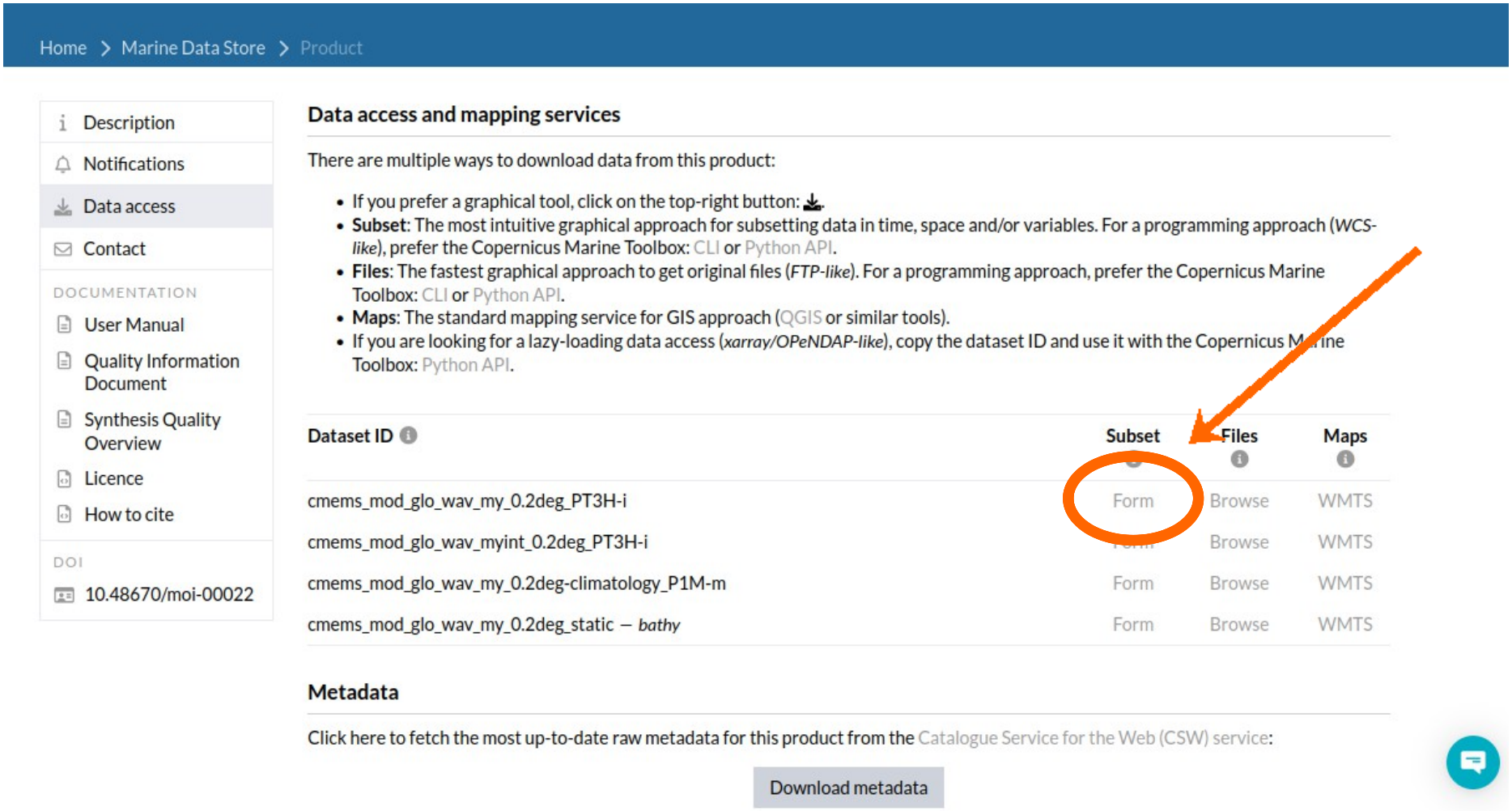
Downloading wave data from Copernicus

Data Access



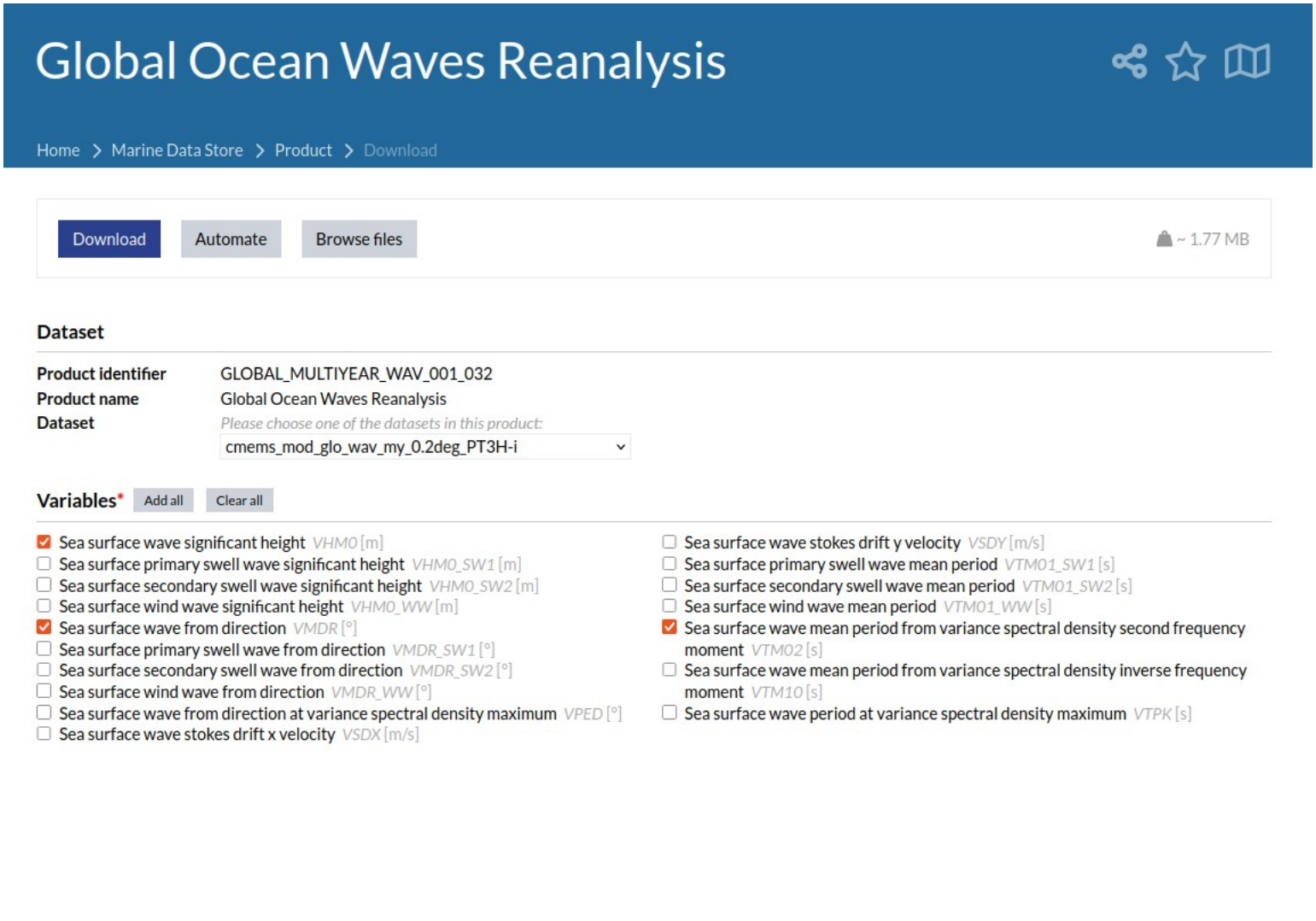
Downloading wave data from Copernicus

Data Access



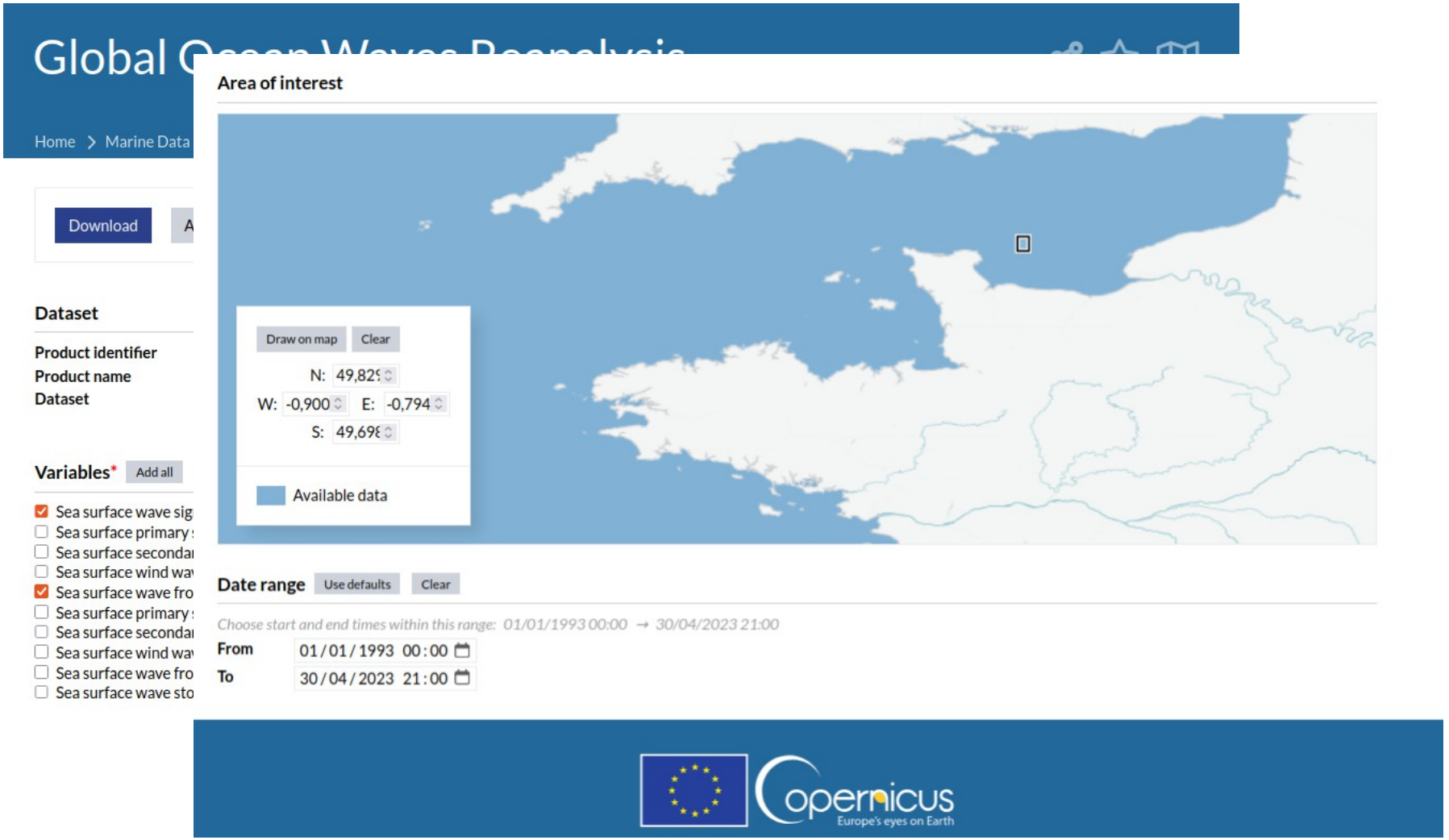
Downloading wave data from Copernicus

Data Access



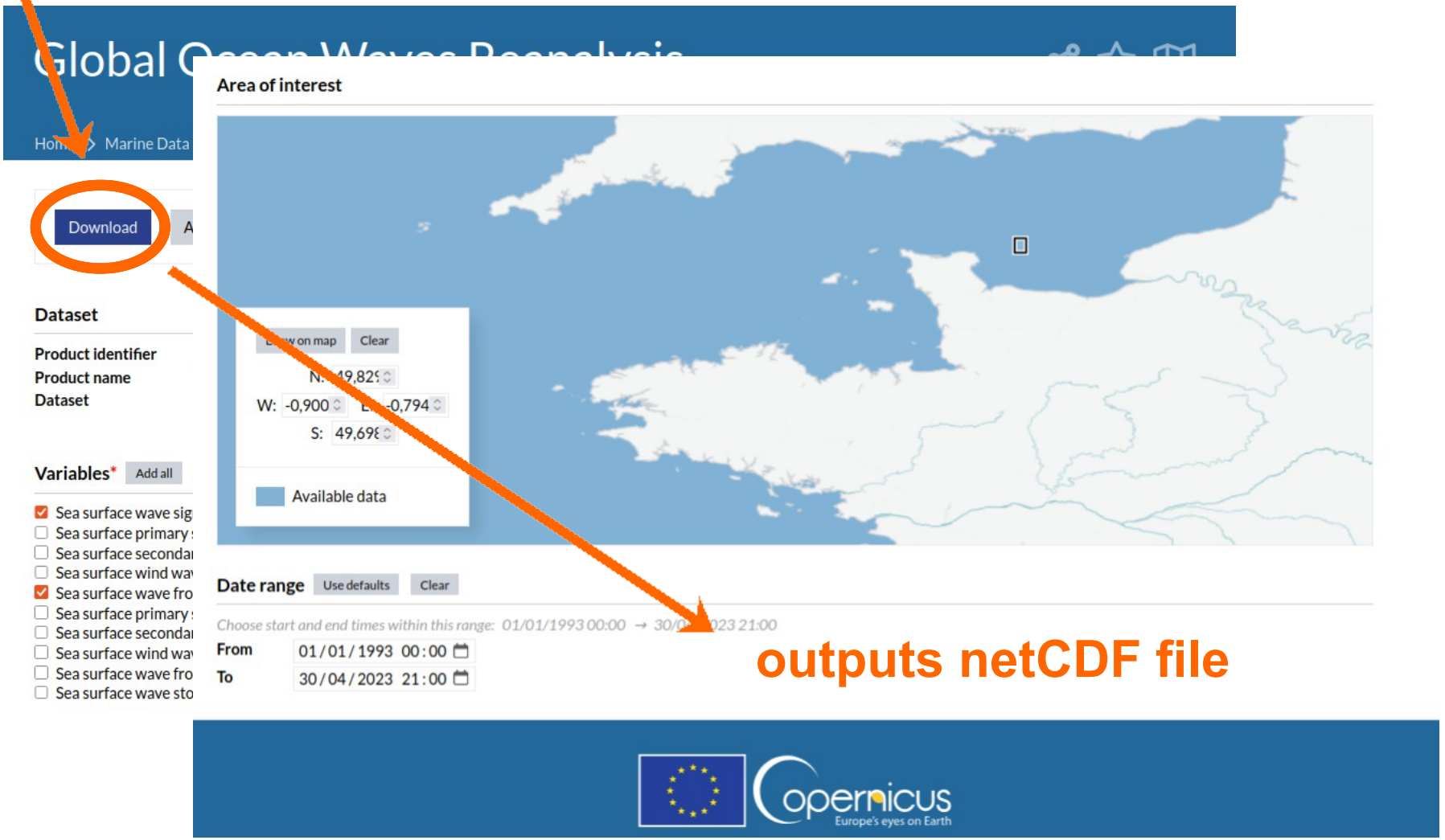
Downloading wave data from Copernicus

Data Access



Downloading wave data from Copernicus

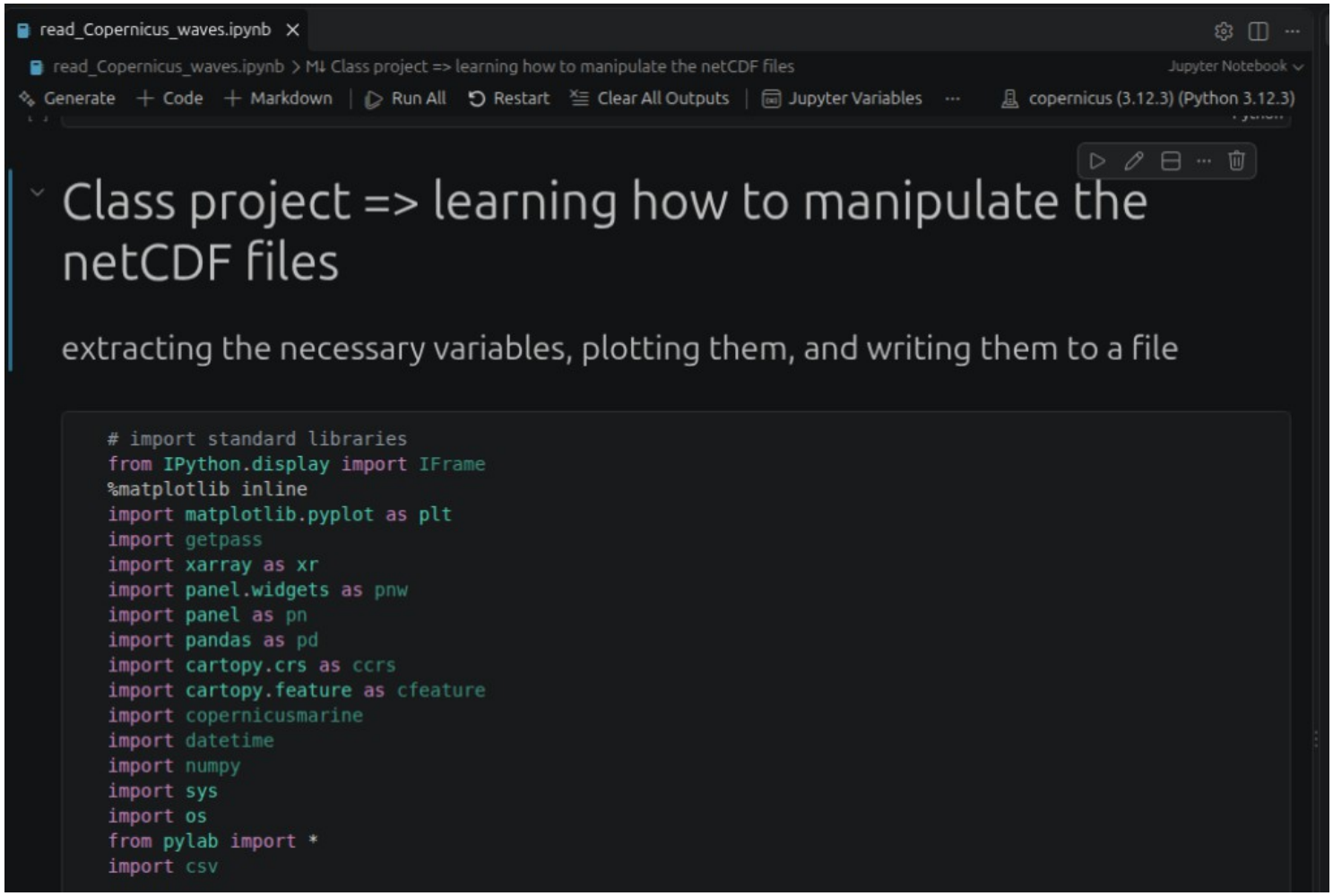
Data Access



Manipulating the wave data

Starting with tools developped by Copernicus...

a jupyter-notebook to get started with the data



Manipulating the wave data

Starting with tools developped by Copernicus...
Choose between installing with pip / mamba / ?

First create a directory where you will be working. Then, before running pip, create a virutal environment (here called copernicus) in the directory that you want to install and work with this virtual environment:

>> Copernicus: python3 -m venv copernicus
>> source copernicus/bin/activate

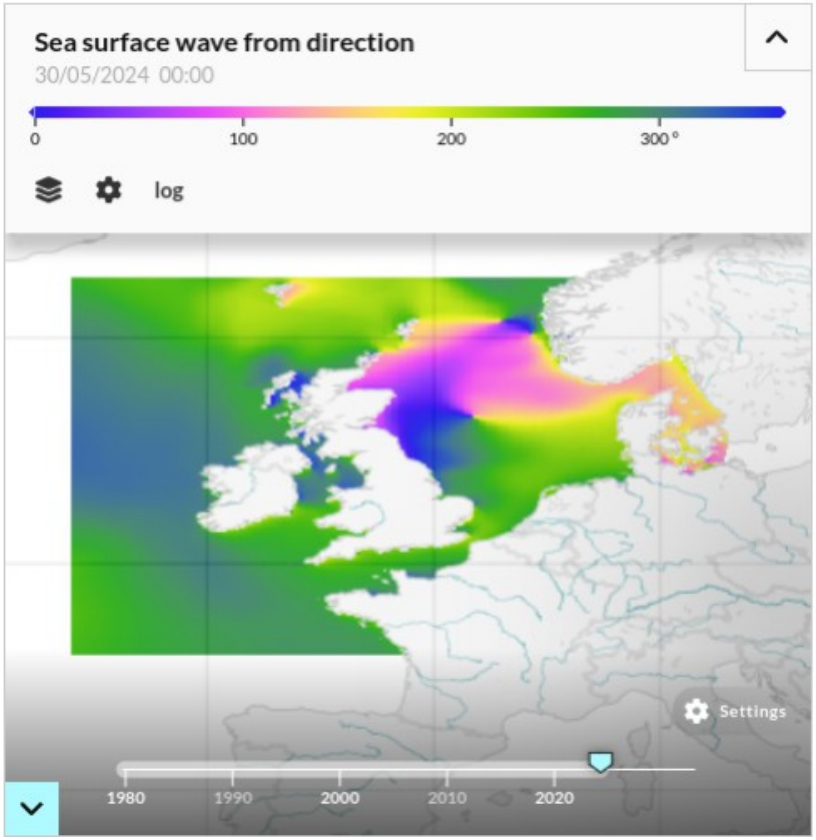
Then you can start installing the necessary packages
>> Copernicus: pip install ipkernel
>> Copernicus: python3 -m pip install copernicusmarine
>> Copernicus: pip install panel, matplotlib, cartopy, netCDF4, wavetools

To test the setup and get started with your analyses, run the example python notebook provided in the cloud with the example provided :

read_Copernicus_waves.ipynb

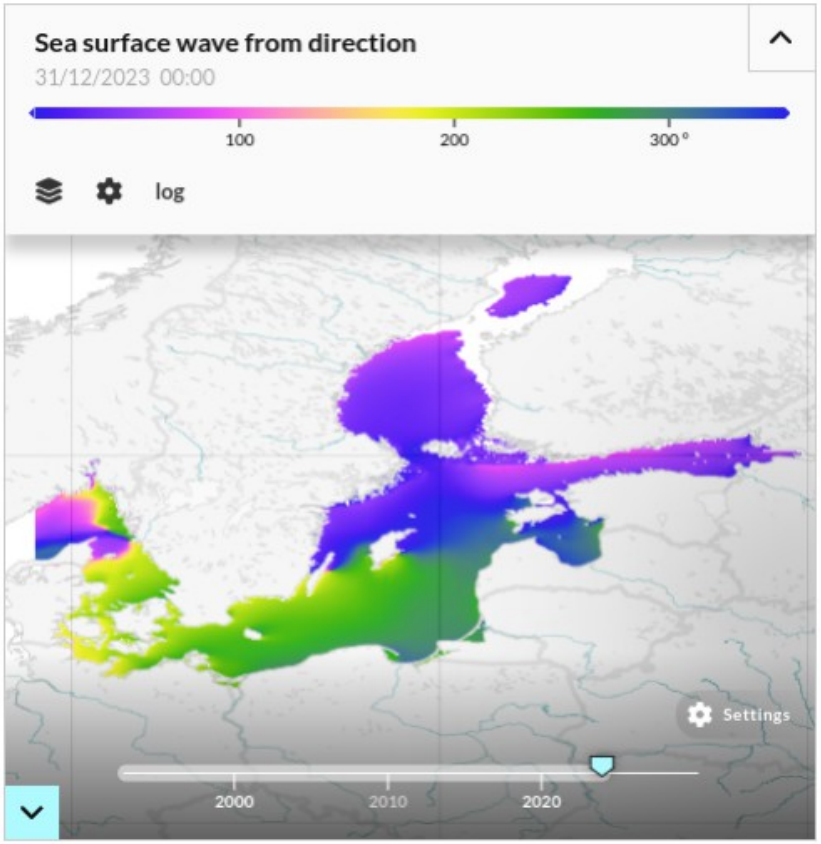
Downloading wave data in your study sites

Atlantic- European North West Shelf- Wave Physics Reanalysis



https://data.marine.copernicus.eu/product/NWSHELF_REANALYSIS_WAV_004_015/description?view=-&option=-&product_id=-

Baltic Sea Wave Hindcast



https://data.marine.copernicus.eu/product/BALTICSEA_REANALYSIS_WAV_003_015/description